WORKED SOLUTION

STEP III 2005, Question 14

Probability and Statistics • Continuous Random Variables

STEP III 2005, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q14

The key idea

Everything here runs on the one integral we are allowed to quote, so the work is in spotting which m and n each integral needs. The substitution $x = k^2/u$ swaps the interval $[0, k]$ with $[k, \infty)$ while leaving the integrand's shape unchanged, and that symmetry pins down the median at once. For the time $T = s/V$, the same reciprocal trick shows that T has a density of exactly the same family as V, with k replaced by s/k. After that, the medians and means can be read straight off the results for V.

Solution

The constant C

Take $m = a$ and $n = 2a$ in the given result. This is allowed because a is a positive integer and $2a \geq a$. Then $n + 2 = 2a + 2$ and $n - m + 1 = a + 1$, so

$$\int_0^\infty \frac{x^a}{(x+k)^{2a+2}}\,\mathrm{d}x = \frac{a!\,a!}{(2a+1)!\,k^{a+1}}.$$

A density integrates to 1 over its range, so

$$1 = Ck^{a+1} \cdot \frac{a!\,a!}{(2a+1)!\,k^{a+1}} \qquad \Longrightarrow \qquad C = \frac{(2a+1)!}{a!\,a!},$$

as required.

The substitution and the median

In the left-hand integral, put $x = k^2/u$, so that $\mathrm{d}x = -\dfrac{k^2}{u^2}\,\mathrm{d}u$. As x runs from 0 to v, u runs from ∞ down to k^2/v. Also

$$x + k = \frac{k^2}{u} + k = \frac{k(u+k)}{u},$$

so

$$\frac{x^a}{(x+k)^{2a+2}} = \frac{k^{2a}/u^a}{k^{2a+2}(u+k)^{2a+2}/u^{2a+2}} = \frac{u^{a+2}}{k^2\,(u+k)^{2a+2}}.$$

Multiplying by $\mathrm{d}x$ cancels the k^2/u^2, and the minus sign reverses the limits:

$$\int_0^v \frac{x^a}{(x+k)^{2a+2}}\,\mathrm{d}x = \int_\infty^{k^2/v} \frac{u^a}{(u+k)^{2a+2}}\,(-\mathrm{d}u) = \int_{k^2/v}^\infty \frac{u^a}{(u+k)^{2a+2}}\,\mathrm{d}u,$$

which is the required identity.

Ask yourself Which value of v makes this identity say something about the median?

Choose $v = k$, so that $k^2/v = k$ as well. After multiplying both sides by Ck^{a+1}, the identity becomes

$$\begin{aligned}\mathrm{P}(V < k) &= \int_0^k \mathrm{f}(x)\,\mathrm{d}x \\ &= \int_k^\infty \mathrm{f}(u)\,\mathrm{d}u = \mathrm{P}(V > k).\end{aligned}$$

V is continuous, so $\mathrm{P}(V = k) = 0$ and these two probabilities add to 1. Each is therefore $\frac{1}{2}$, and the median of V is k.

The expected value of V

$$\mathrm{E}(V) = Ck^{a+1}\int_0^\infty \frac{x^{a+1}}{(x+k)^{2a+2}}\,\mathrm{d}x.$$

Now take $m = a+1$ and $n = 2a$. This needs $2a \geq a+1$, which holds because $a \geq 1$. Then $n - m + 1 = a$, and

$$\mathrm{E}(V) = \frac{(2a+1)!}{a!\,a!}\,k^{a+1}\cdot\frac{(a+1)!\,(a-1)!}{(2a+1)!\,k^a} = k\cdot\frac{(a+1)!}{a!}\cdot\frac{(a-1)!}{a!} = \frac{(a+1)k}{a}.$$

The distribution of T

Speeds are positive, so for $t > 0$ we can multiply through by $V/t > 0$ without reversing the inequality:

$$T < t \iff \frac{s}{V} < t \iff V > \frac{s}{t}.$$

Hence

$$\mathrm{P}(T < t) = \mathrm{P}\Big(V > \frac{s}{t}\Big) = \int_{s/t}^\infty \frac{Ck^{a+1}x^a}{(x+k)^{2a+2}}\,\mathrm{d}x, \qquad (*)$$

as required.

Now substitute $u = s/x$ in $(*)$, so that $x = s/u$ and $\mathrm{d}x = -\dfrac{s}{u^2}\,\mathrm{d}u$. As x runs from s/t to ∞, u runs from t down to 0. Since $x + k = (s+ku)/u$,

$$\mathrm{P}(T < t) = \int_0^t Ck^{a+1}\,\frac{s^a/u^a}{(s+ku)^{2a+2}/u^{2a+2}}\cdot\frac{s}{u^2}\,\mathrm{d}u = \int_0^t \frac{Ck^{a+1}s^{a+1}\,u^a}{(s+ku)^{2a+2}}\,\mathrm{d}u.$$

This holds for every $t > 0$, and $\mathrm{P}(T < t) = 0$ for $t \le 0$. So differentiating with respect to t gives the density of T:

$$\mathrm{g}(t) = \frac{Ck^{a+1}s^{a+1}\,t^a}{(s+kt)^{2a+2}} \qquad (t > 0).$$

To compare this with f, take out k^{2a+2} from the denominator and write $K = s/k$:

$$\mathrm{g}(t) = \frac{C\,(s/k)^{a+1}\,t^a}{(t+s/k)^{2a+2}} = \frac{CK^{a+1}t^a}{(t+K)^{2a+2}}.$$

This is exactly the density f with k replaced by $K = s/k$. So T belongs to the same family of distributions as V.

Medians and means

Because T has the same form of distribution with parameter $K = s/k$, the earlier results apply directly:

$$\text{median of } T = \frac{s}{k}, \qquad \mathrm{E}(T) = \frac{(a+1)}{a} \cdot \frac{s}{k}.$$

The product of the median time and the median speed is therefore

$$\frac{s}{k} \cdot k = s,$$

which is the distance, as claimed. The product of the expected time and the expected speed is

$$\mathrm{E}(T)\,\mathrm{E}(V) = \frac{(a+1)s}{ak} \cdot \frac{(a+1)k}{a} = s\Big(1 + \frac{1}{a}\Big)^2 > s,$$

since a is a positive integer, so $1 + 1/a > 1$. This completes the question.

Checks

- For $a = 1$, $k = 1$ we get $C = 3!/1 = 6$, and numerical integration of $6x/(x+1)^4$ over $[0, \infty)$ gives 1.000000. Over $[0, 1]$ it gives 0.500000, confirming the median.
- Numerical quadrature for several pairs, such as $a = 2$, $k = 0.7$, gives $\mathrm{E}(V) = 1.05000 = \frac{3}{2}(0.7)$. With $s = 3.2$ it gives $\mathrm{E}(T)\mathrm{E}(V) = 7.20000 = 3.2 \times \frac{9}{4}$.
- A simulation of 2×10^5 speeds with $a = 2$, $k = s = 1$ gave median speed $\times$ median time ≈ 1.000001, but mean speed ≈ 1.504 (theory: 1.5).
- The median product equalling s can be seen without any formulae. $T = s/V$ is a decreasing function of V, so the median of T is s divided by the median of V.

Common slips

- Forgetting to check $n \ge m$ when quoting the given integral. For $\mathrm{E}(V)$ it needs $a \ge 1$, which is exactly why a must be a positive integer. For larger moments the formula would eventually fail, and indeed $\mathrm{E}(V^{a+1})$ is infinite.

- Losing the factor s/t^2 (the chain rule) when differentiating $(*)$ to find the density of T.
- Assuming $\mathrm{E}(s/V) = s/\mathrm{E}(V)$. The whole point of the last part is that this is false.

Beyond STEP

The final inequality is an instance of Jensen's inequality: $1/v$ is convex for $v > 0$, so $\mathrm{E}(1/V) \geq 1/\mathrm{E}(V)$, with equality only if V is constant. Medians, by contrast, pass through any monotonic function. The substitution $V/(V+k)$ turns V into a $\mathrm{Beta}(a+1, a+1)$ variable, which is symmetric about $\frac{1}{2}$, and this is another way to see why the median is k. Blitzstein and Hwang, *Introduction to Probability*, covers Jensen's inequality, change of variables and the Beta family very readably.

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